

Autonomous Systems and Mobile Robots – Tutorial

Introduction and Organisational



Autonomous Intelligent Systems Group



Malte Schilling

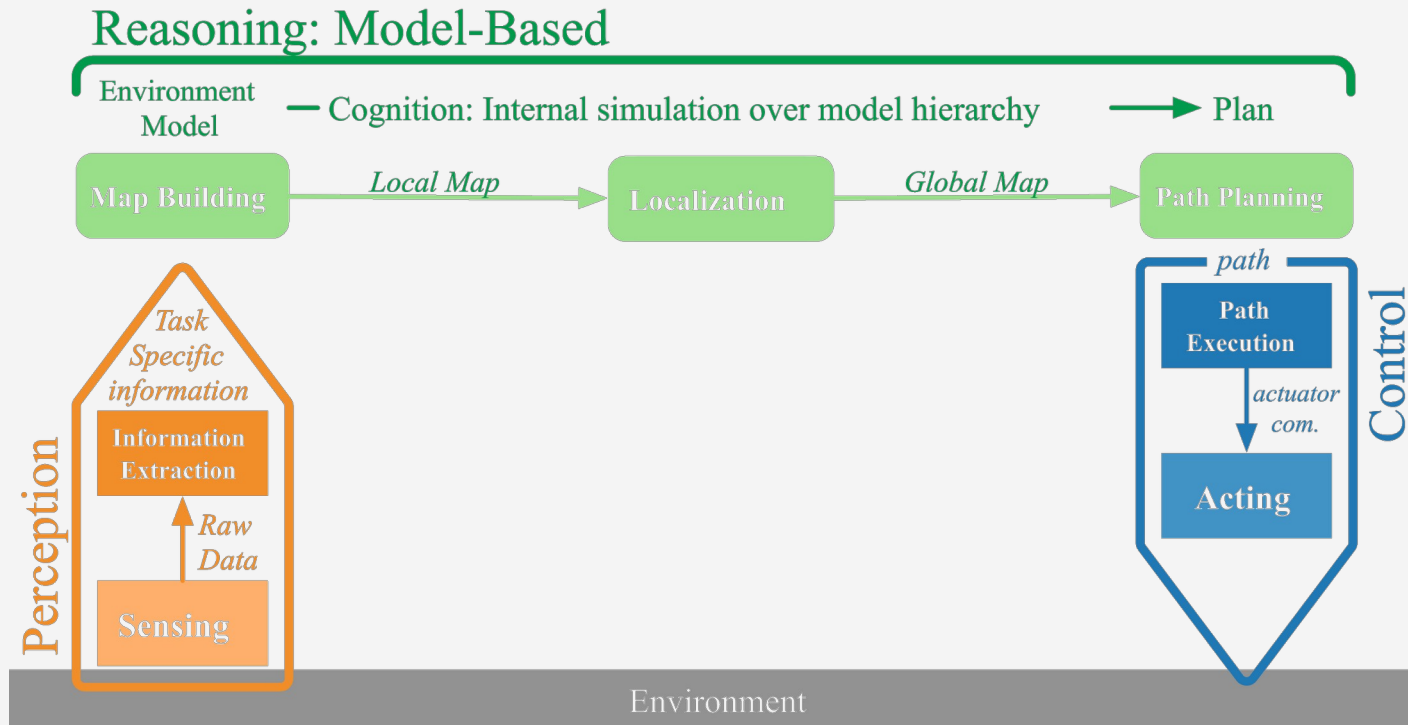


Simon Neumeyer



Janosch Bajorath

Perceive – Reason – Act



Unitree GO1

6-Axis Inertial Measurement Unit

3-Axis Accelerometer
3-Axis Gyroscope

Compute Power

1 Raspberry 4B+
3 Nvidia Jetson Nano (Xavier)

5x Stereo Depth Cameras

10x Fish-eye Cameras

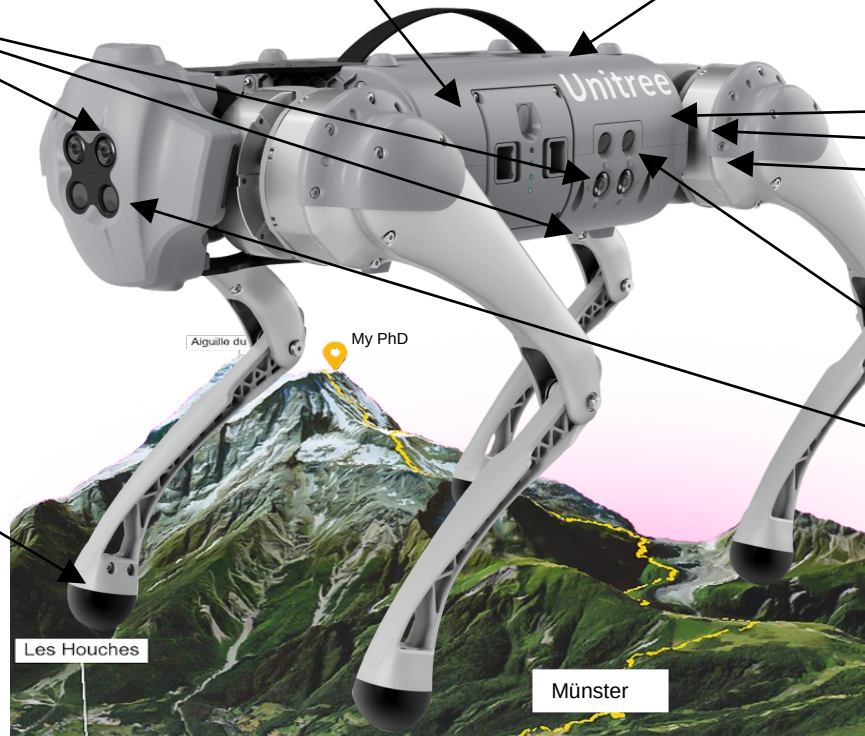
12x Joint Actuators

Hip Rotation: $-40^{\circ} - 40^{\circ}$
Hip Flexion: $-218^{\circ} - 45^{\circ}$
Knee Flexion: $24^{\circ} - 132^{\circ}$

Foot Contact Sensor

3x Hypersonic Distance Sensors

Weight: 12.5 kg (no battery)
LxWxH: 0.645 x 0.28 x 0.29 m
Speed: up to 3.5 m/s
Motor: 23.7 Nm
Climb: 35°
Battery: 1-2 hrs



Robust and Adaptive Quadruped Locomotion

Training Iteration 0

Learning a Quadruped
Control Policy using Deep
Reinforcement Learning

Robust and Adaptive Quadruped Locomotion



Research Focus

- 1) High-dimensional embodied control
- 2) Robustness across environments
- 3) Adaptation and behavioural flexibility
- 4) Structured internal representations

Robust and Adaptive Quadruped Locomotion

Live Demo
-
Unitree GO1

Structure and Appointments

Voluntary Tutorial Sessions'

1. Reinforce knowledge presented during lectures.
2. Hands-on work in Python and ROS2 on small example problems.
3. New Jupyter notebooks or ROS2 Packages with tasks each week.

When: Fri. 12:15 - 13:45

Where: M5, Einsteinstr. 62

Structure and Appointments*

Week	Date	PRA Theme	Tutorial Focus	Actual Lecture
0	14.4.	-	Onboarding, PRA overview, Python setup	Nr. 1 - Einführung (Autonomie, PRA)
1	21.4.	Act: control	ROS2 basics and control with quadruped demo	Nr. 2 - Actuation (Open/Closed Loop, PID)
2	28.4.	Act: communication	ROS2 communication patterns	Nr. 3 - Control (Locomotion, wheeled drives, motion models)
3	5.5.	Reason: robot body	Robot modelling (URDF, TF, Gazebo) - intuitive frames	Nr. 4 - Hardware: Sensors
4	12.5.	Perceive: sensors	Sensors and uncertainty	Nr. 5 - Kinematics (Transformation, FK)
5	19.5.	Integrate PRA	MiniLab 1 briefing	Nr. 6 - Kinematics (3D Rotation, IK, Jacobian, DH)
	26.5.	-	<i>Pfingsten</i>	<i>Pfingsten</i>
6	2.6.	Reason: FK/IK	Manipulator kinematics (FK/IK);	Nr. 7 - Architekturen (Layered, Subsumption)
7	9.6.	Perceive + Reason	Localisation and state estimation	Nr. 8 - Localization (+ Path Planning)
8	16.6.	Reason: planning	MiniLab 2 briefing	Nr. 9 - SLAM
9	23.6.	Full stack	Navigation (and path planning)	Nr. 10 - Navigation
10	30.6.	-	MiniLab 1 and 2 recap	Nr. 11 - Planning
11	7.7.	-	Optional Recap Session	Nr. 12 - Lernen in Robotern
-	14.7.	-	(no tutorial)	Nr. 13 - Zusammenfassung

*preliminary

Structure and Appointments

Mini Labs

- Two mandatory mini-lab exercises must be submitted in teams of 2–3 students.
- Each mini-lab must reach > 50 % to qualify for the final exam.

Collaboration Policy:

1. Discuss ideas together with other groups, but write and code your own solutions.
2. Do not share or publish your code or write-ups online.

Structure and Appointments

Group Allocation

- Mini-Lab submissions
- Compute resource approval

Email to j.bajorath@uni-muenster.de

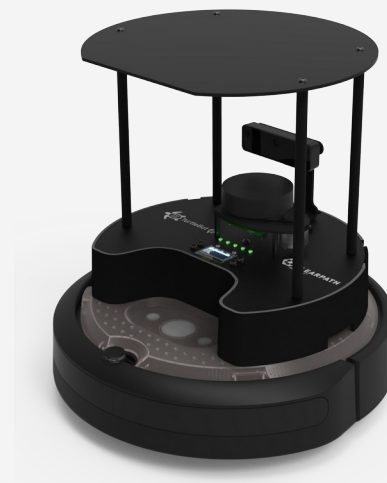
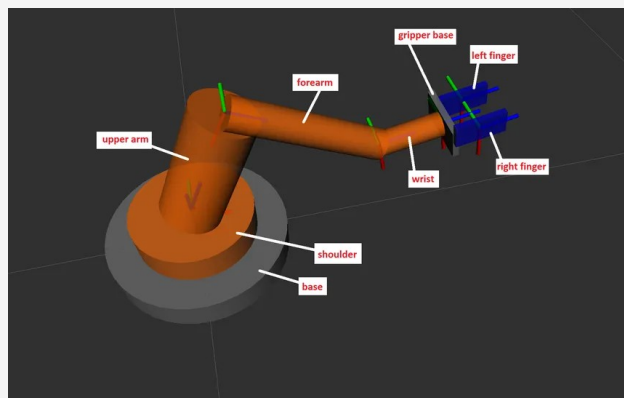
Deadline: 25.04.2026

Robot Platforms



Unitree GO1

Simple Robot Arm



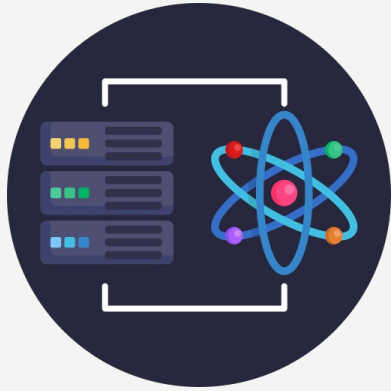
Turtlebot (or similar)

Python Environment Options

Live Demo
-
Local Setup and University Infrastructure

PRA example Exercise

main



exercise.ipynb

helper (if needed)



Resources

JupyterHub NRW

<https://www.jupyterhub.nrw/>

ASMR GIT Project

<https://zivgitlab.uni-muenster.de/ai-systems/teaching/public/26-ss/asmr>

ASMR Tutorial

<https://zivgitlab.uni-muenster.de/ai-systems/teaching/public/26-ss/asmr/tutorial>